

Figure 7.47 Myopic eye and correction

can see as shown in Figure 7.47(b). To overcome this difficulty, the virtual image of the object at infinity should be formed at a distance x from the eye using a correcting lens as shown in Figure 7.47(c).

The focal length of the correcting lens for a myopic eye can be calculated using the lens equation (6.63).

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

Here, $u = -\infty$, $v = -x$. Substituting these values in the lens equation gives,

$$\frac{1}{f}=\frac{1}{-x}-\frac{1}{-\infty}$$

Focal length f of the correcting lens is,

$$f = -x \qquad (7.83)$$

The negative sign in the above result suggests that the correcting lens should be a concave lens. Basically, the concave lens slightly diverges the parallel rays from infinity and makes them fall at the retina.

7.6.7.2 Farsightedness (*hypermetropia*)

A person suffering from *farsightedness* (or) *hypermetropia* (or) *hyperopia* cannot see closer object clearly. It occurs when the eye lens has long focal length (or) shortening of the eyeball than usual. The closest distance for clear vision for these people is appreciably more than 25 cm. Thus, reading books (or) viewing smaller things held in the hands is difficult for them. This kind of farsightedness arising due to aging is called *presbyopia* as the aged people cannot strain their eye more to reduce the focal length of the eye lens.

The rays coming from the object at near point get focused beyond the retina as shown in Figure 7.48(a). But, these persons can see objects which are at a distance more than 25 cm. Let y be the minimum distance from the eye beyond which a person with farsightedness can see as shown in Figure 7.48(b). To overcome this difficulty, a virtual image of the object at y should be formed at a distance of 25 cm (near point) from the eye using a correcting lens as shown in Figure 7.48(c).

The focal length of the correcting lens for a hypermetropic eye can be calculated using the lens equation (6.63).

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

Here, u = –25 cm, $v = -y$. Substituting these values in the lens equation gives,

$$\frac{1}{f}=\frac{1}{-y}-\frac{1}{-25\text{ cm}}$$

Simplifying the above equation for f gives,

$$\frac{1}{f}=\frac{1}{25\text{ cm}}-\frac{1}{y}=\frac{y-25\text{ cm}}{y\times 25\text{ cm}}$$

$$f=\frac{y\times 25\text{ cm}}{y-25\text{ cm}} \qquad (7.84)$$

The focal length calculated using above formula will be positive as y is always

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